

that are relatively cheap to make and easy to operate.

Photonic equivalents

A PIC or integrated optical circuit is a device that integrates multiple (at least two) photonic functions and as such is analogous to an electronic IC. The major difference between the two is that, a PIC provides functionality for information signals imposed on optical wavelengths, typically in the visible spectrum or near infrared (IR) 850nm to 1650nm. Unlike electronic integration, where silicon is the dominant material, system PICs have been fabricated from a variety of material systems, including electro-optic crystals such as lithium-niobate, silica on silicon, silicon on insulator, various polymers and semiconductor materials, which are used to make semiconductor lasers such as gallium-arsenide (GaAs) and indium-phosphide (InP).

Different material systems are used because each provides different advantages and limitations, depending on the function to be integrated. Photonic equivalents of these devices have been equally challenging to develop and are widely used to manipulate and control the signals in optical fibres. But, it is fair to say that these have yet to reach their full potential.

Photonic crystals are useful because their optical properties are determined by the physical geometry, size of the waveguide and so on. This allows these to be precisely tuned to carry only certain wavelengths of light.

PIC is a breakthrough technology as it uses photons (smallest unit of light) as data carriers instead of electrons (smallest unit of electricity) used in electronic ICs. As light travels at very high speeds, PIC technology is widely used to transfer huge amounts of data at a very high speed. Thus, PIC based products are primarily deployed in the field of optical-fibre communications.

PICs market is growing at a phenomenal rate as it provides significant improvements in system size, power consumption, reliability and cost. Development of silicon photonics

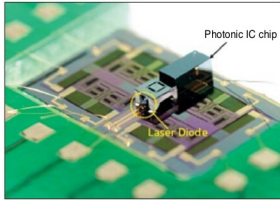


Fig. 2: A photonic integrated circuit (Image courtesy: www.research.a-star.edu.sg)

technology has helped in large-scale manufacturing of PICs at a low cost.

Also, current leading players have developed monolithically-integrated InP based PICs that can integrate more than 600 components/functions on a single chip. Thus, there is a huge competition in the market as each player is trying to innovate PIC based products, which would be able to integrate large amounts of functions/components at a low cost.

Fabrication techniques

PIC fabrication techniques are similar to those used in electronic ICs, in which photolithography is used to pattern wafers for etching and material deposition.

Unlike electronics, where the primary device is the transistor, there is no single dominant device. The range of devices required on a chip includes low-loss interconnect waveguides, power splitters, optical amplifiers, optical modulators, filters, lasers and detectors. These devices require a variety of different materials and fabrication techniques, making it difficult to realise all of these on a single chip. Newer techniques using resonant photonic interferometry are making way for UV LEDs to be used for optical computing requirements with much cheaper costs, leading the way to petahertz (PHz) consumer electronics.

Development status

The most commercially-utilised material platform for PICs is InP, which

allows for the integration of various optically-active and passive functions on the same chip.

Initial examples of PICs were simple two-section distributed Bragg reflector lasers, consisting of two independently-controlled device sections: a gain section and a DBR mirror section.

Consequently, all modern monolithic tunable lasers, widely tunable

lasers, externally-modulated lasers and transmitters, and integrated receivers, among others are examples of PICs. Current state-of-the-art devices integrate hundreds of functions onto a single chip.

Some technologists report the development of the first integrated optical circuit with a photonic crystal capable of both localising and interfacing atoms with guided photons in the device. The device is a photonic crystal made of silicon-nitride that acts as a waveguide for laser light to carry light tuned to certain atomic transitions in cesium. When a cesium atom absorbs and scatters these wavelengths, the process generates forces that can be used to trap and manipulate the atom. The photonic crystal is integrated into a system that provides a ready supply of cesium atoms, and the result is an IC capable of manipulating individual cesium atoms.

Recently, the field of plasmonics, which exploits surface plasmons generated when photons hit a metal structure, has opened up the real possibility that photonic circuits could duplicate what electronic ICs do. Previously, photonic circuits were just too large to be functional because of their need to accommodate different wavelengths of light. Despite several advances, plasmons still lost energy too quickly, which reduced the distance these could travel. Now, researchers have developed a solution to this issue by combining graphene and